# Markscheme

Mathematics: analysis and approaches
Practice question bank

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**81. (a)**

$\$3000$ is invested at an annual interest rate of $r\%$, compounded annually. After 6 years, the investment is worth $\$4200$.

Find the value of $r$, correct to 3 significant figures.

$$3000(1+\tfrac{r}{100})^6=4200$$ *(M1)*

$$(1+\tfrac{r}{100})^6=1.4$$ **A1**

$$1+\tfrac{r}{100}=1.4^{\frac{1}{6}}$$ *(M1)*

$$r\approx 5.77$$ **A1**

**[5 marks]**